Multimodal AI Teaching Copilot: Live Slide Generation from Lecture Speech

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***Abstract*—Prepared slides cannot show what a teacher adds while explaining, and a large language model (LLM) that regenerates slides from a growing transcript rewrites what students are reading. This paper presents a teaching copilot that builds a structured, continuity-aware slide deck on the classroom projector from the teacher's live speech. Locally transcribed speech is sent in bounded units to an LLM that only returns a schema-validated interpretation; a deterministic planner turns it into extend-only slide operations that cannot rewrite what students have read, and a transcript-coverage guard restores facts the LLM omits. On 36 held-out lectures (409 min) on a laptop with a 4 GB GPU, the median speech-to-slide delay was 11.2 s. On an annotated subset of 18 lectures, slide changes matched topic boundaries with precision 0.78, 86.6 % of key facts reached a slide, 0.9 % of statements were unsupported by the speech, and five teachers rated the slides 3.9–4.2 of 5. On the same units, an LLM-controlled baseline rewrote or removed 38.2 % of the statements students had already seen, against 2.0 % (lower in all 18 lectures, $p < 0.001$), and a replay ablation showed that opening a slide per concept lowers precision to 0.37.**

***Index Terms*—Automatic slide generation, educational technology, large language models, live systems, speech recognition, human-in-the-loop**

I. Introduction

Slides prepared before a lecture fix its order in advance, so the examples and answers a teacher adds while explaining never reach the screen. Learners understand better from words with organized pictures than from words alone [1], if the visuals add no extraneous load [2], [3], but no teacher can author such visuals while teaching.

Existing tools solve only parts of this problem: live captions show verbatim text without the structure of the lesson [4]; slide generators work offline from a finished document or prompt [5]–[11]; and a large language model (LLM) [12], [13] applied to a growing transcript rewrites slides while students read them and adds untaught content [14]. Three problems remain open: (P1) projected material does not follow what is taught; (P2) live speech is not turned into a structured visual lesson; and (P3) LLM output is neither stable nor faithful to the teacher's words.

This paper presents the Multimodal AI Teaching Copilot, an implemented system that addresses all three: spoken input becomes visual output (typed layouts, typeset formulas, diagrams and retrieved images) on the projector while the teacher speaks. The contributions are: (1) incremental slide generation from live speech; (2) an extend-only update policy: the LLM only labels bounded, schema-validated units, and a deterministic planner enforces slide operations that add to the deck and cannot rewrite what students have read; (3) a transcript-coverage guard that restores facts the LLM omits, and a formula guard; and (4) a held-out evaluation with annotation, teacher ratings, a replay ablation and a paired comparison with an LLM-controlled baseline.

II. Related Work

A. Multimedia Learning and Lecture Speech

Mayer showed that words with pictures are understood better than words alone [1]; Sweller's cognitive load theory adds that working memory is limited [2], [3], and unrelated pictures lowered performance in PowerPoint lectures [15]; hence short typed layouts and an image only for a concrete taught object.

Whisper by Radford *et al.* [16], run with 8-bit weights by faster-whisper [17], and Silero VAD [18] form the speech front end. Kushalnagar *et al.* showed that verbatim real-time captions are hard for students to follow [4], and Gonzalez *et al.* found that LLM summaries of recorded lectures improved quiz scores [19]; both produce text, not a structured visual lesson (P2).

B. Segmentation, Slide Generation and LLMs

TextTiling by Hearst places subtopic boundaries where the similarity of adjacent blocks drops [20]; Sentence-BERT by Reimers and Gurevych compares sentence embeddings by cosine similarity [21], and Wang *et al.* distilled it into MiniLM [22]. These methods segment a complete transcript offline; here the similarity drop is one online signal, evaluated with the measure of Beeferman *et al.* [23].

PPSGen selects sentences of a paper for slides by integer linear programming [7], D2S summarizes document text for each slide title [6], and DOC2PPT generates slides with text and figures from scientific documents [5]. With LLMs, Xie *et al.* ground educational slides in retrieved sources [8], SlideBot adds multi-agent planning [9], PPTAgent edits reference slides [10], and AutoPresent writes slides as code [11]. All build a whole deck offline from written input; a lecture arrives incrementally as noisy speech (P1, P3).

Kasneci *et al.* stressed teacher oversight of LLMs in education [12], so the teacher overrides every automatic decision. Ji *et al.* surveyed hallucination [14], and Willard and Louf constrained generation to a JSON schema [24], which fixes the format but not the facts, so the output is also checked against the transcript; generated quizzes follow the review of Kurdi *et al.* [25].

TABLE I
COMPARISON OF THE PROPOSED SYSTEM WITH PRIOR SYSTEMS

System	Input	Produced	Topic structure	Faithfulness to source	Slide sequence
Live captions [4], [16]	Speech	Live	None (text stream)	Verbatim text	None
Lecture summaries [19]	Recorded lecture	Offline	None (summary)	LLM summary	None
PPSGen, D2S, DOC2PPT [5]–[7]	Paper / document	Offline	Document sections	Extracted from document	Whole deck at once
LLM slide generators [8]–[11]	Topic, document	Offline	LLM outline / plan	Retrieved sources or none	Whole deck at once
Proposed system	**Live speech**	**During lecture**	**Online topic / subtopic**	**Coverage + formula guards**	**Extend-only + teacher**

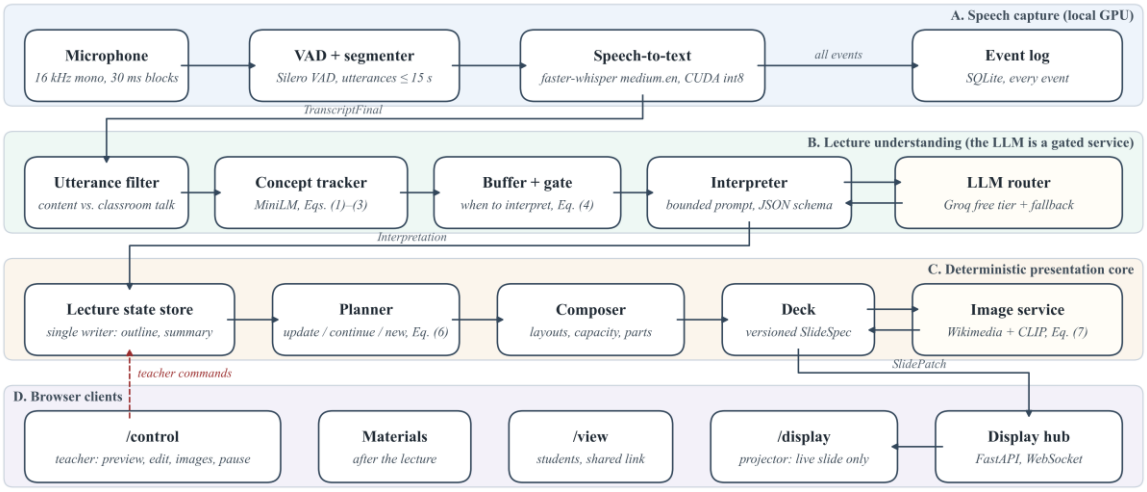


Fig. 1. Implemented architecture. Lanes A–C run in one Python process; only the interpreter calls the LLM; browsers get slide patches and send commands.

Among the systems in Table I, none combines streaming speech input with incremental decisions about which slide to change while students read; this is the gap addressed here.

III. METHODOLOGY

A. System Overview

A lecture is processed as a stream through six stages (Fig. 1): speech capture and recognition; filtering and concept tracking; a gate that groups finished sentences into units; bounded LLM interpretation of each unit; a deterministic planner and composer that turn it into slide operations and layouts; and delivery to the projector, the teacher and students. Four rules guide the design: the LLM supplies labels but never executes a slide operation, it never sees the whole transcript, a new concept is not by itself a new slide, and user interfaces send commands instead of changing the state.

All components run in one Python *asyncio* process and exchange typed events; every event is logged in SQLite, from which each lecture is replayed for evaluation and materials. Recognition runs on the GPU, MiniLM and CLIP on the CPU (ONNX), and the LLM on the Groq free tier (openai/gpt-oss-120b [13]; qwen/qwen3.8-27b [26] when that quota is spent; Oct. 2026), as a 7B-class local model does not fit beside the speech model in 4 GB. Of the 1,619 held-out units, 71.6 % were interpreted by the first model, 24.0 % by the second and 4.4 % by the deterministic fallback.

B. Speech Capture and Concept Tracking

Silero VAD [18] segments 16 kHz audio (utterances up to 15 s), and faster-whisper [17] with Whisper medium.en [16] (int8, beam size 1, keyword prompt) transcribes them. A deterministic filter removes classroom management, meta talk and fillers. Each content utterance is embedded with MiniLM [22]; with the L2-normalized embedding e_t and concept centroid c_{t-1}, the shift score is the cosine distance $s_t = 1 - e_t \cdot c_{t-1}$ used with Sentence-BERT [21], and the centroid is the exponentially weighted mean $c_t = \mathrm{norm}((1-\alpha)c_{t-1} + \alpha e_t)$ with $\alpha = 0.35$, raised to 0.6 at a boundary so it re-seeds on the new concept. Following the similarity-drop criterion of TextTiling [20], a boundary is signalled when

$$b_t = \kappa_t \vee (s_t \geq \tau), \tau = 0.75 \quad (1)$$

where κ_t marks a cue phrase ("moving on to", "next"); τ is high because shift scores within one facet were 0.3–0.6.

C. Gate and Token Budget

Content accumulates in a buffer until a rule-based gate fires. With p_t the silence after a finished sentence, w_t the buffered words and a_t the age of the oldest pending line (s),

$$G_t = (p_t \geq 1.2 \wedge w_t \geq 6) \vee w_t \geq 40 \vee b_t \vee a_t \geq 12 \quad (2)$$

with an 8 s floor between calls, set by the free quota (200,000 tokens per key and model per day).

D. Bounded Interpretation and Transcript-Grounding Safeguards

The prompt holds the subject, level, an outline (≤ 150 tokens), the current slide with numbered items, a rolling summary (≤ 120 tokens) and the new lines, within 3,000 tokens (temperature 0.2). The LLM returns JSON with topic, subtopic, a relation (same concept, elaboration, sub-concept, sibling, new topic, digression), typed content, an image hint

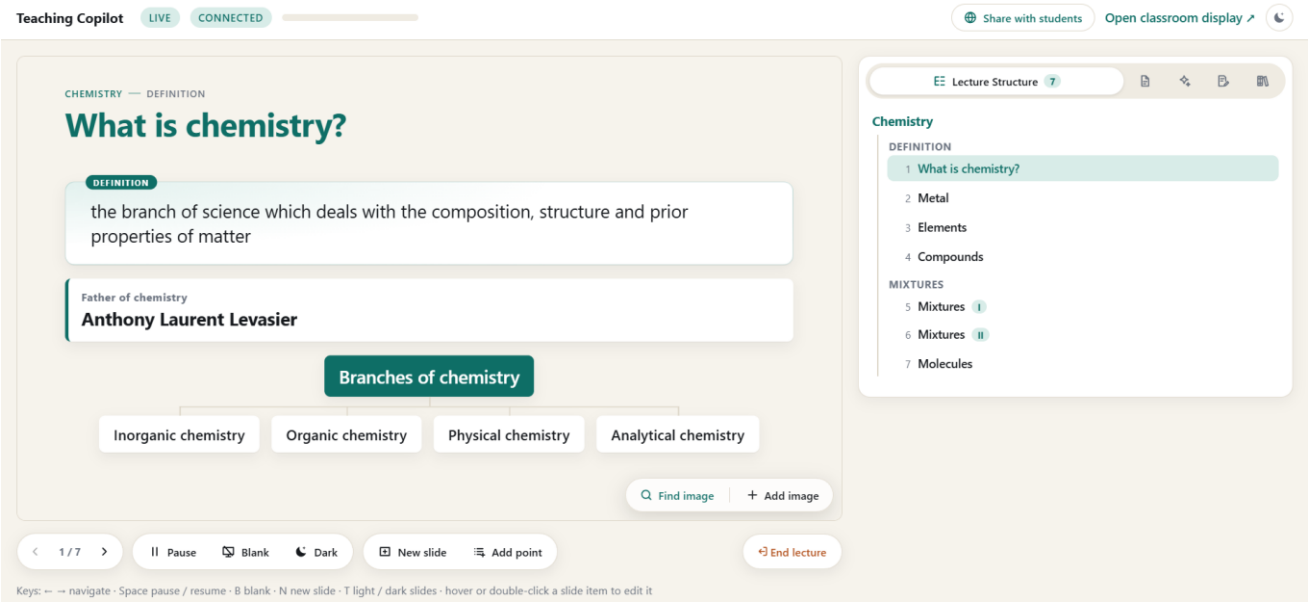


Fig. 2. Teacher control view (/control) in a live chemistry lecture: the projected slide as an editable preview with image tools and lecture controls (left) and the lecture structure tree built by the system, topic → facet → slides, with overflow parts I and II (right).

and *concerns* about possibly wrong statements. Pydantic validates it; after one failed repair, a deterministic fallback shows the spoken sentences as points. A coverage guard compares every content sentence u (≥ 6 words) with the output O; with $W(\cdot)$ the set of word stems and a recall-style overlap as in ROUGE-1 [27],

$$|W(u) \cap W(O)| \leq \tfrac{1}{3}|W(u)| \Rightarrow u \text{ is shown as said} \quad (3)$$

so a fact the model left out still reaches the slide. A formula guard flags changed formula tokens, and a correction is applied only above a confidence of 0.75 (factual) or 0.4 (mis-hearing); otherwise the slide shows what was said and the teacher gets a card with both versions. These checks keep slides faithful to the transcript; they do not prove a statement correct.

E. Slide Planner

The planner turns each interpretation into one operation: *update* the working slide, *continue* on a new slide of the same topic, open a *new* topic slide, *retitle*, or nothing. None of them lets the LLM rewrite or delete a shown item: shown text changes only when the composer merges an elaboration into the item it elaborates, a mis-hearing correction is applied, or the teacher edits it. A topic change must be confirmed, because a single unit may be misread:

$$C_n = B_n \vee \left(\bar{s}_n^{\max} \geq \tau\right) \vee \left(T_n \equiv \tilde{T}_{n-1}\right) \quad (4)$$

where B_n is true if a boundary (1) occurred in unit n, $\bar{s}_n^{\max}$ is its largest shift score smoothed over two utterances, T_n its topic and $\tilde{T}_{n-1}$ an unconfirmed candidate from the previous unit. Unconfirmed content is shown tentatively and moved once confirmed. A minimum dwell of 15 s (6 s for the next part of a full slide) prevents rapid changes, near-duplicates (similarity ≥ 0.85) are merged, and teacher edits are never overwritten. All thresholds were set by hand on D1 and are tested by ablation in Section IV.

F. Composer and Image Retrieval

The composer maps items to definition cards, key points, fact tiles, process flows, comparison tables, timelines, cause-effect chains, classification trees and formulas; overflow moves to a further part (I, II, …) instead of shrinking text, and spoken mathematics is typeset with KaTeX. For a concrete object the LLM suggests a query; an openly licensed Wikimedia image (source, author and licence kept) is placed only if CLIP's zero-shot similarity [28] ranks it above generic distractor prompts ("a logo", "a map") and the query matches the slide text.

G. Live Display and Teacher Control

A FastAPI server pushes versioned slide patches over WebSocket to /display (projector), /control (teacher) and /view (students, shared link). The control view (Fig. 2) adds editing, image search and upload, pause, blank screen, mistake cards, the running transcript and the lecture structure tree.

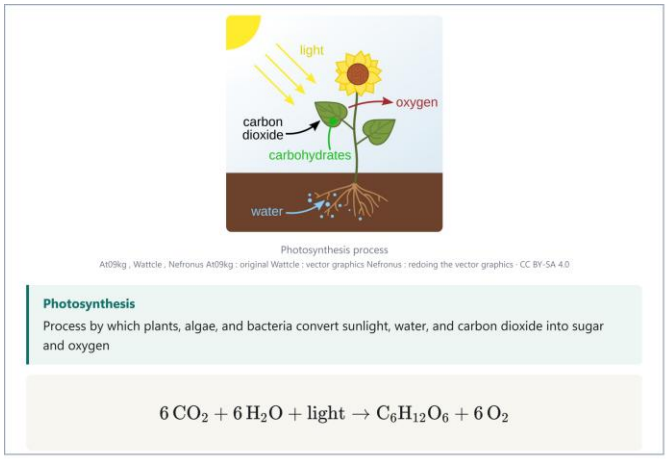
H. Lecture Materials

After the lecture the event log is reused: on request it becomes study notes, an assignment (Fig. 3), a PPTX deck, or a live quiz or test answered through the shared link, all from the taught content and edited by the teacher; the teacher's own notes can be indexed locally to spell terms. These features are not evaluated here.

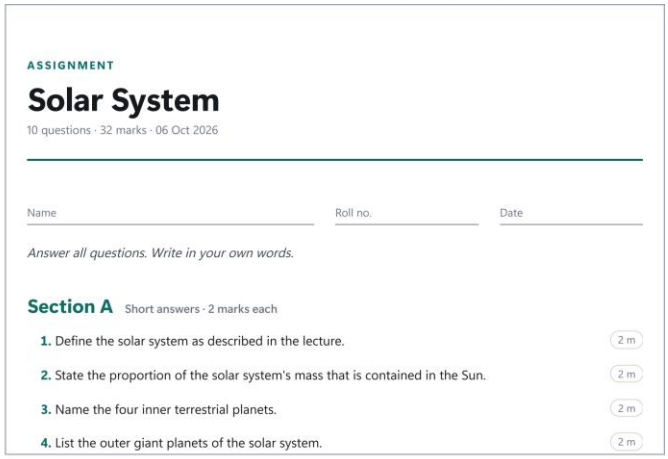
IV. Experimental Evaluation

A. Setup and Data

All runs used a laptop with an RTX 3050 GPU (4 GB) and 16 GB RAM. D1 is six lectures spoken by the authors (114.1 min). The held-out set D2 (Table II) has 30 lectures spoken live, unscripted, by six speakers into the laptop microphone



(a) Study notes (PDF, page 3)



(b) Assignment (PDF)

Fig. 3. Materials generated from the event log of a recorded D1 lecture (not evaluated): (a) a study-notes page; (b) the start of the assignment.

(Recorded, physics to economics) and six 10-min public lectures played in as audio in real time (Public): MIT OpenCourseWare 7.016, 5.111, 18.01 and 8.01 (CC BY-NC-SA) and NPTEL Computer Networks and History of Economic Theory (CC BY-SA). The speakers consented to the recording. Against MIT's human captions of the 7.016 lecture (1,257 words), the live transcript has a word error rate of 7.1 %. Fallback units (a quota spent or a provider failure) still produced slides. Slides were not edited during a test; only two items were deleted in all of D2. Three populations are evaluated: all 36 lectures for latency and operation, an annotated subset of 18 (12 Recorded, 6 Public; 191 min) for accuracy and the ablation, and 30 sampled slides for teacher ratings.

TABLE II
EVALUATION DATA AND LATENCY (SESSION LOGS)

Set	Lect.	Min	Units (fb)	Slides	STT	LLM	E2E (p95)
D1	6	114.1	389 (45)	163	0.66	1.35	12.5 (36.8)
D2 Recorded	30	349.2	1275 (59)	453	0.71	1.92	10.1 (23.2)
D2 Public	6	60.0	344 (13)	52	0.60	2.23	17.1 (93.4)
D2 all	**36**	**409.2**	**1619 (72)**	**505**	**0.68**	**1.96**	**11.2 (53.0)**

fb: fallback units; STT, LLM: median per utterance / call (s); E2E: median (95th percentile) from the end of a spoken line to its display (s); Public includes web4, web6 (provider outages).

Annotation. Two authors (A and B) independently annotated the 18 lectures, chosen to mix subjects and speakers with 6–17 min each. Before seeing the slides, each marked topic boundaries (where a teacher would start a new board) and key facts in the transcript, then judged whether each key fact was on a slide and whether each slide statement was said and correct; scores are the mean of A and B, without adjudication. A system boundary is the utterance that opened a new content slide. Boundaries are scored by precision, recall and F1 within ±1 utterance, averaged per lecture, and by P_k [23], the probability that two utterances half a segment apart are wrongly split or joined; agreement is Cohen's κ [29] on the scale of Landis and Koch [30] and, for skewed labels, Gwet's AC1 [31]. Intervals are 95 % bootstrap intervals over lectures. *Ratings.* Five teachers (T1–T5), not involved in development, rated 30 random D2 slides, each shown with what was said, from 1 to 5 on four criteria; their consistency is the ICC(2,1) of Shrout and Fleiss [32].

B. Topic Boundaries and Content Accuracy

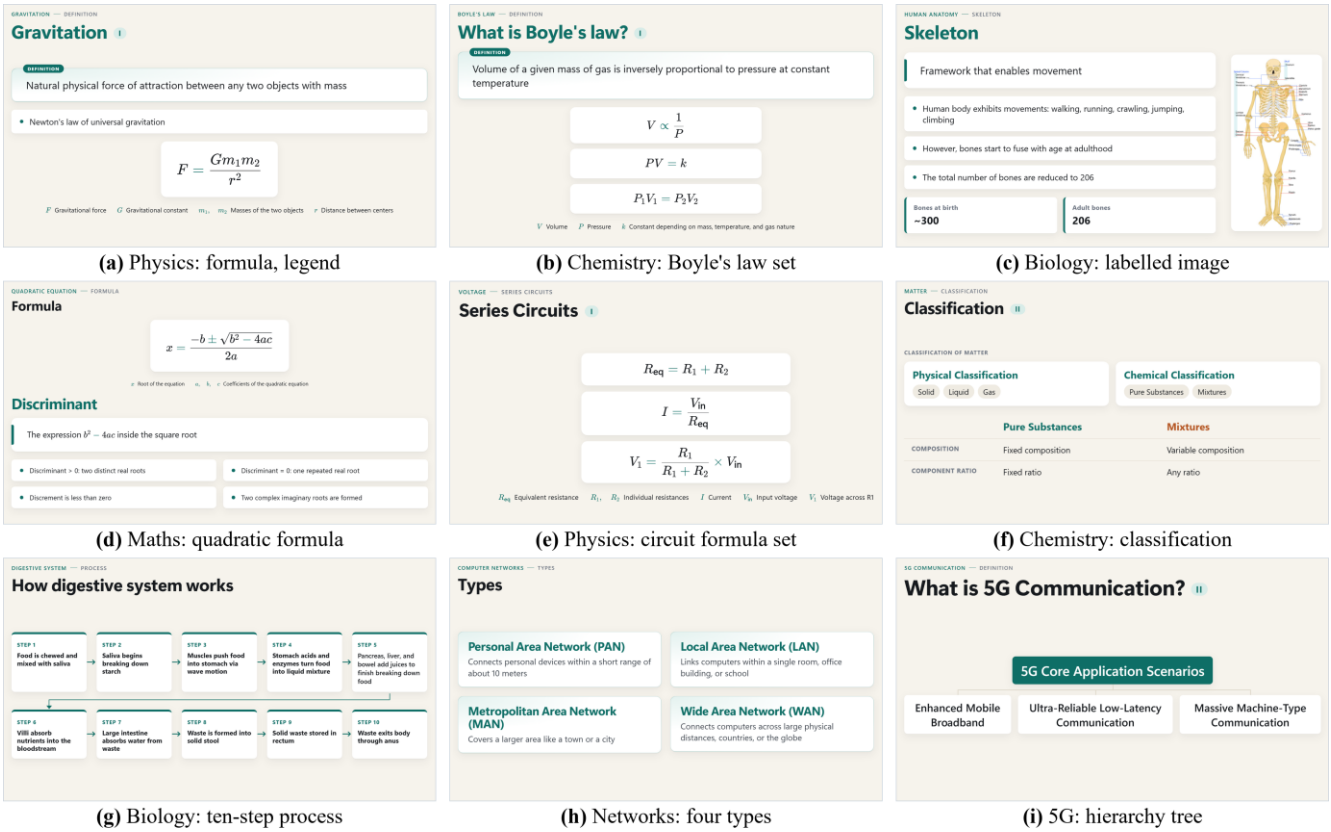
Slides change where the topic changes (Table III): boundary precision is 0.78 (95 % interval 0.68–0.88), so a projected slide is rarely replaced within a topic, and P_k is 0.32. Recall is 0.42 (0.30–0.55): a new concept is not a new slide, so the planner merges into the live slide sub-points that annotators marked as boundaries, most often in the Public lectures (recall 0.19); Section IV-C shows this is a trade-off. Of 669 key facts, 86.6 % (84.6–88.5 %) reached a slide (partly shown counts ½), and only 0.9 % (0.2–1.8 %) of 2,504 statement judgments were not supported by the speech. The coverage guard (3) is needed for this: 60 of the covered facts reached a slide only through a sentence it restored, so without it coverage would be 77.7 %; the guard adds 9.0 points (6.4–11.7). Of the 3.7 % marked wrong, 41 of 79 were also marked as said (a slip or mis-hearing): slides follow the speech, and corrections go to the teacher's concern cards; transcript support is therefore not factual correctness. Agreement is substantial for boundaries (κ = 0.72 [30]) and high for "supported" (88.8 %, AC1 0.88); κ is only 0.07 there because nearly all statements were supported (the prevalence paradox [33]).

TABLE III
ANNOTATED ACCURACY AND TEACHER RATINGS (18 D2 LECTURES)

Measure	Recorded	Public	All
Boundary precision / recall	0.81 / 0.54	0.73 / 0.19	0.78 / 0.42
Boundary F1 / P_k	0.59 / 0.29	0.29 / 0.37	0.49 / 0.32
Key-fact coverage	88.0 %	84.0 %	86.6 %
without coverage guard	81.9 %	69.1 %	77.7 %
Unsupported / wrong	0.9 / 3.6 %	0.9 / 4.0 %	0.9 / 3.7 %
κ boundaries (agreement)	0.81 (94 %)	0.56 (90 %)	0.72 (92 %)
AC1 supported (agreement)	0.89 (90 %)	0.84 (85 %)	0.88 (89 %)
Rating: correct. / relevance	4.20 / 3.94	4.06 / 3.80	4.15 / 3.89
Rating: readab. / usefulness	4.12 / 3.93	4.26 / 3.76	4.17 / 3.87

Means over annotators A and B; boundaries ±1 utterance; unsupported / wrong: share of slide statements; ratings: mean of T1–T5 (1–5; 20 Recorded, 10 Public slides).

Test29, interpreted entirely by the fallback (quota spent), covered 83.1 % of key facts with 0.7 % unsupported



(a) Physics: formula, legend **(b)** Chemistry: Boyle's law set **(c)** Biology: labelled image

(d) Maths: quadratic formula **(e)** Physics: circuit formula set **(f)** Chemistry: classification

(g) Biology: ten-step process **(h)** Networks: four types **(i)** 5G: hierarchy tree

Fig. 4. Projected D1 slides: (c), (d), (f) are screenshots of the application; the others re-render recorded LLM interpretations (no new calls). Text is as spoken, e.g. the slip "Discrement" in (d).

statements but rarely opened a slide (recall 0.04) and used three layouts, 11 of its 13 slides plain points, against 12 layouts and 34 % plain points in the other lectures: the fallback preserves content, the LLM adds the structure. The 17 LLM lectures reach precision 0.77 and recall 0.45. The teachers rated the slides 3.87–4.17 of 5 (ICC(2,1) 0.45–0.56, moderate). In Fig. 4 the layout follows the kind of content, not the subject.

C. Ablation of the Planner Rules

Table IV replays the recorded concept signals and LLM answers of the 18 lectures, without new LLM calls, through the planner with one rule changed; the unchanged replay reproduces the live decks exactly, and a variant approximates its live behaviour.

The continuity rule carries the result: if each concept opens a slide, the deck changes 3.2 times per minute, 200 slides are replaced within 10 s, and precision and P_k drop to 0.37 and 0.47. The dwell time delays a slide change rather than cancelling it, so changes per minute stay at 1.15, but slides replaced within 10 s fall from 25 to 16 without loss of accuracy; a dwell of up to 20 s leaves precision and recall unchanged, while 45 s lowers them to 0.67 and 0.32. The topic confirmation brought no measurable benefit on D2 (precision 0.79 without it); it remains as a safeguard. When the LLM's labels decide directly but slides are still only extended, the boundaries are similar: the stability comes from the extend-only operations, not from the label source; the planner enforces them.

TABLE IV
PLANNER ABLATION BY REPLAY (18 ANNOTATED D2 LECTURES)

Variant	P	R	P_k	Chg./min	< 10 s
Full planner (= live)	**0.78**	0.42	**0.32**	**1.15**	16
Without dwell time	0.78	0.42	0.32	1.15	25
Without topic confirmation	0.79	0.45	0.31	1.12	10
LLM labels decide directly	0.78	0.46	0.30	1.17	21
Each concept a new slide	0.37	0.86	0.47	3.18	200

P, R: boundary precision, recall (mean of A and B); Chg./min: slide changes per minute; < 10 s: slides replaced within 10 s of being shown.

D. Comparison With an LLM-Controlled Baseline

The ablation keeps the extend-only operations. Without them, an LLM-controlled baseline receives, for the same 850 units of the 18 annotated lectures, the slide it showed last and the last 60 s of transcript, and returns the slide to show, without planner, guards or composer (same two models, 0.5 M tokens). It placed slide changes about as well (Table V; precision 0.72 vs. 0.78, $p = 0.34$) but rewrote or removed 1,439 of the 3,763 statements students had already seen (38.2 %), against 91 of 4,515 (2.0 %), and 65.8 % of what it showed was gone from its final deck, against 6.2 %. The difference holds in all 18 lectures: the mean per-lecture gap in rewrites is 39 points (95 % interval 32–47; exact paired sign-flip permutation test, $p < 0.001$). Of the proposed system's 91 changes, 24 extended a statement, 7 were teacher edits and 60 composer merges.

TABLE V
PROPOSED SYSTEM VS. LLM-CONTROLLED BASELINE (SAME D2 UNITS)

System	Slides/min	Rewritten	Missing	P	P_k
Proposed	1.25	**2.0 %**	**6.2 %**	**0.78**	**0.32**
LLM baseline	0.79	38.2 %	65.8 %	0.72	0.36

Rewritten: statements of a shown slide that its next version rewrote or removed; missing: distinct statements shown but absent from the final deck; 18 annotated lectures.

E. End-to-End Latency

Over all 36 D2 lectures (Table II), planning and composing take 2 ms. From the end of a spoken line to its content on the display the median is 11.2 s, of which 7.6 s is the deliberate wait in the gate for a complete idea; the system is live, not hard real-time. In web4 and web6 the free provider failed (timeouts, HTTP 429), units queued, and the 95th percentile rose to 53.0 s; without these two lectures the median is 10.2 s and the 95th percentile 23.4 s.

V. DISCUSSION AND CONCLUSION

The Multimodal AI Teaching Copilot builds a continuity-aware slide deck from live speech, with the LLM as a bounded interpreter and extend-only operations enforced by a deterministic planner: on held-out lectures it kept 86.6 % of key facts with 0.9 % unsupported statements, and an LLM in control rewrote 38.2 % of shown statements, against 2.0 %. Limitations: the annotators were authors (marking boundaries and facts before seeing the slides); recall is moderate by design; an append-only baseline prompt was not tested; the word error rate comes from the one lecture with human captions; the free provider caused outages; and without a classroom study no claim is made about learning. Anonymized annotations, ratings and evaluation scripts are available from the authors on request.

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